PhD Progress book

Andrew Habrich

2023-12-18

Contents

1 About **5**

1.1 Usage 5

1.2 Research topic 6

2 Meta-analysis **7**

Missed connections in the city: A meta-analysis of landscape connectivity factors affecting biodiversity in urban ecosystems 7

2.1 Methods 8

2.2 Screening process 9

2.3 Initial results 10

2.4 Interactive map 14

2.5 Data extraction 16

2.6 Next steps: 17

3 ebird checklist diversity in Canadian cities **19**

3.1 Research question 19

3.2 ebird dataset download 20

3.3 Methods 20

3.4 Ottawa - Gatineau 21

3.5 Interactive map 22

3.6 Next steps: 22

4 ebfly data in canadian cities **27**

4.1 ebutterfly data download 27

4.2 Checklists filtered by geography 28

4.3 Checklists by year of submission 28

4.4 Ottawa - Gatineau 29

4.5 Interactive map 31

4.6 Next steps: 31

Chapter 1

About

This is a book document meant to track progress on my PhD projects in a reproducible and organized report. In this document you'll find tables, figures, and descriptive code for state of progress for each of the projects. Most of the work for these projects is done with R, and the reports (including this one) are generated with R Markdown.

The functional code and folder directories for each of the R projects can be found on my github page: https://github.com/drewhabrich. Note though, that not all the README.md files are updated yet (work-in-progress).

1.1 Usage

The directory on the left is clickable and should help with navigation. You can also move through the document using the < and > keys and search for keywords with the magnifying glass on the top left of the page.

There are 2 versions of this report that you can view: the **first** is this webpage, hosted on my github page, and **second** as a PDF of the website generated directly from R. Note though that the PDF version is functional, but the layout and figure sizings are a bit buggy right now. The PDF file can be downloaded by clicking the little pdf icon on the toolbar at the top of the page.

One last thing that you can do on the webpage is **add annotations and highlight text** with the use of the 'Hypothes.is' platform (https://hypothes.is/). You can make a free account and add comments, annotations, and questions directly on the webpage by clicking the < at the top right side of the page, or by highlighting text and clicking the "Annotate" button. Posting an annotation to the page will upload the comment directly to the hypothesis page.

1.2 Research topic

Landscape connectivity and bird and butterfly diversity in the urban ecosystems:

Conceptually simple, landscape connectivity is described as the degree to which the landscape configuration affects movement of organisms (Tischendorf and Fahrig 2000; Taylor et al. 1993). Connectivity has become central to many biodiversity conservation efforts (Rudnick et al. 2012; Keeley, Beier, and Jenness 2021), including city-specific planning (Nilon et al. 2017), as well as a topic of interest in international agreements (Convention on Biological Diversity 2022).

Despite the simplicity of the concept, in practice, landscape connectivity is measured using a broad range of statistical and theoretical approaches, all of which have different assumptions, interpretations and implications for landscape management (Correa Ayram et al. 2016; Fletcher et al. 2019; Fahrig et al. 2021).

Recent narrative reviews on the impacts of landscape connectivity on urban biodiversity have identified inconsistencies in the literature, particularly around the clarity of statements describing *a priori* hyptheses, and definitions of 'connectivity' and 'urban' (LaPoint et al. 2015; Lookingbill et al. 2022). It follows that inconsistent or unclear use of connectivity concepts throughout the literature may constrain comparability between studies and complicate interpretation, limiting our capability to identify the real impact of connectivity on biodiversity

As such, there remains a need to improve understanding of effects urban landscape connectivity, and a necessity to validate these connectivity models with consistent biological data across multiple cities. The projects in my thesis aim to provide clarity on this topic, and to develop transparent, reproducible frameworks to incorporate available biodiversity data and remote-sensing imagery to validate and facilitate landscape connectivity modelling in urban areas.

Chapter 2

Meta-analysis

The following R packages (and their versions) were used to generate this report.

```
## kableExtra         sf    mapview  patchwork  rphylopic synthesisr CiteSource
##    "1.3.4"   "1.0-14"   "2.11.2"    "1.1.3"    "1.2.2"    "0.3.0"    "0.0.1"
##  lubridate    forcats    stringr      dplyr      purrr      readr      tidyr
##    "1.9.3"    "1.0.0"    "1.5.0"    "1.1.3"    "1.0.2"    "2.1.4"    "1.3.0"
##     tibble    ggplot2  tidyverse
##    "3.2.1"    "3.4.4"    "2.0.0"
```

Missed connections in the city: A meta-analysis of landscape connectivity factors affecting biodiversity in urban ecosystems

With the ongoing and expanding use of connectivity metrics in conservation planning, there is considerable value in disentangling the true effect of connectivity on biodiversity from methodological and statistical artifacts to inform sustainable urban development. I intend to review the urban connectivity literature to answer the following research questions:

1. What is the overall direction and magnitude of its effect on urban biodiversity?
2. What variation/heterogeneity among studies can be explained by biological factors (taxonomy, life history traits etc.) and anthropogenic factors (geographic location, city size, human population density, etc.)
3. To what extent have urban connectivity models been validated with independently obtained biological data in cities?

4. How does the theoretical approach/connectivity metric used depend on the taxa studied or spatial scale, and does it affect the overall direction/magnitude of the effect?

2.1 Methods

2.1.1 Search strategy

I used an intial set of search terms that encapsulate the three main concepts of the review: (1) urban environments, (2) connectivity types, (3) biodiversity outcomes.

Table 2.1: Table 1: Search term concepts

Concept	Search string
Urban environments	urban OR "urban area" *OR "urban landscape**" OR city OR cities OR town*
Connectivity concepts/types	connectivity OR "landscape connectivity" OR "functional connectivity" OR "structural connectivity" OR "habitat connectivity" OR "ecological connectivity" OR "habitat network" *OR "ecological network"* OR corridor* OR "habitat corridor" *OR "ecological corridor"* OR least-cost OR "least cost" OR "circuit theory" OR circuit-theory OR "landscape resistance*" OR "landscape permeability"
Biodiversity outcome	situp* OR occurr* OR abundan* OR "species richness" OR diversity OR "speci* response" *OR "gene* diversity" OR "gene flow" OR "speci* distribut" *OR "speci* densit" *OR "speci* composit" *OR "speci* assemblage" *OR "communit* composit" *OR "communit* assemblage" *OR "popul* persist" *OR "popul* response" *OR "distribut* pattern" *OR "spatial* distribut" *OR movement* OR dispersal

A preliminary search was then conducted on Jan. 25th 2023 using the search strings in the Web of Science Core Collection. The search results were downloaded as [.bib] or [.ris] files, depending on availability. From my original literature review and this initial search, I identified a set of articles that captured all three relevant concepts. Next, using a keyword co-occurrence networks I used the search results to find additional search terms to improve the comprehensiveness of the search, using the 'litsearchr' package in R (Grames et al. 2019).

After identifying a set of final search strings, I searched databases on Apr. 3rd 2023 for both primary peer-reviewed literature (**Web of Science Core Col-**

Table 2.2: Inclusion criteria used for title/abstract screening

Criteria	Description
1	Does the study mention that the research was conducted in a specific city or urban area, or was studyin
2	Does the study explicitely mention connectivity or describe a predictor variable that could be interprete
3	Does the study measure some estimate of biodiversity (e.g., species richness, index of diversity etc.), abu
4	Is the focal taxa/species terrestrial for some part of their life cycle (includes amphibians but exclude aq

lection, SCOPUS) and unreviewed grey literature (**bioRxiv, ProQuest, and ConservationCorridor.org**), adjusting the search syntax to the specific databases accordingly. Searches were restricted to records in English, though this search limitation may miss regional studies, especially localized reports, but I tried to limit this bias by conducting forward and backward citation searches of key review articles on the topic of urban connectivity using the 'citationchaser' package in R (Haddaway, Grainger, and Gray 2021).

2.2 Screening process

Results from the database searches were downloaded and imported into R to merge and de-duplicate records using the *revtools* package (Westgate 2019), which can flag potential duplicates based on DOI, title, and abstract similarity.

The resulting deduplicated records were then screened systematically at two stages, firstly on the titles/abstracts and secondly on full-text.

2.2.1 Eligibility criteria

For the first Title/Abstract stage I used the following criteria:

Records retained from the title/abstract screening were imported to Zotero to download and manage full-text articles as .pdf files. Depending on institutional access, Zotero (with a plug-in) can automatically bulk download hundreds of articles based on DOI, facilitating management of full texts during the screening efforts.

The full-text screening is documented in an excel spreadsheet with information on the city/urban area, the biodiversity metric(s) used as a response, and the connectivity metric(s) used, with indication of which the specific eligibility criteria that ineligible records violated.

Table 2.3: Inclusion criteria used for full-screening screening potential articles for eligibility

Criteria	Description
1	PDF is inaccessible, through library or other online extraction
2	Document is not in english/Unreliable translation
3	No part of the study is in an 'urban area'
4	Only PART of the study is in an 'urban area', not the main focus of the study
5	No landscape configurational or connectivity metric is used as an explanatory variable
6	The response variable is not related to diversity, abundance, movement, or health of biologi
7	The focal taxa is not terrestrial for at least a part of its life cycle
8	Biological information is not used as a response or to validate the the connectivity model
9	No 'case study' or example given, if the paper was a framework/review/methodology/simul
10	Study reports no relevant extractable information for our study
11	Unclear or missing data; Authors need to be contacted
12	Review paper, browse citations for additional relevant information/articles

2.3 Initial results

Searching multiple databases yielded a more comprehensive search, as shown in the figure below. The overlap of unique records between two major literature databases, Web of Science (WoS) and SCOPUS, was 47% and 72%, respectively of search results found in the other database. Searching multiple databases is critical for a comprehensive review on the research topic!

Expectedly, the most common journals for publication (ignoring BioRxiv) are Landscape and Urban Planning, Urban Ecosystems, PLoS One, and Landscape Ecology.

2.3.1 Title/Abstract screening

Here is a PRISMA diagram for the results of the screening based on the titles/abstracts, with information on the number of records obtained from each of the search databases, records from previous reviews on the topic, and additional citations found through citation networks.

After removing duplicates, I screened **6471** records based on the relevance of the title, abstract, and keywords (if available) using the *metagear* R package (lajeunesse). From these results, I identified 1275 potentially eligible articles for full-review, including 795 flagged as **MAYBE** and 480 flagged as **YES** to for full-text screening. for inclusion.

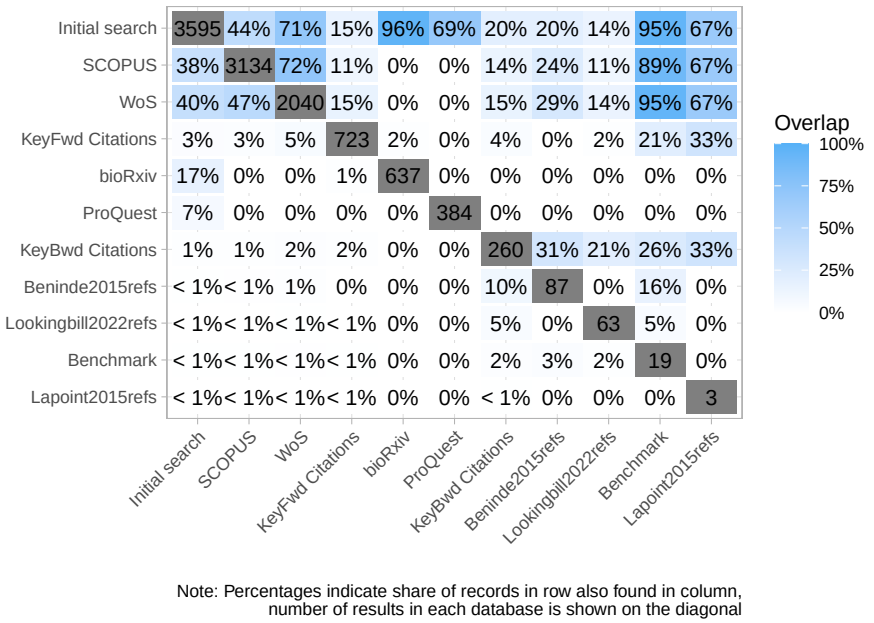


Figure 2.1: Percentage overlap of search results across database sources

2.3.2 Full-text screening

All of the potentially eligible articles were saved as bibliographic references and imported into Zotero. *Most* records were downloaded automatically based on their DOI, but some (mostly student theses) were downloaded manually from their respective repositories.

The following results are just for the published articles that I flagged as **YES** (excluding theses).

For the records rejected at the full-text screening stage, the most common inclusion criteria that were violated were criteria 5 (No landscape configuration/connectivity metric was used as an explanatory variable) and criteria 8 (No biological information was used as a response variable or used to validate the connectivity model) Table 2.3

Research is also not evenly distributed across taxonomic groups, with most work done using birds, arthropods, and mammals as their focal taxa.

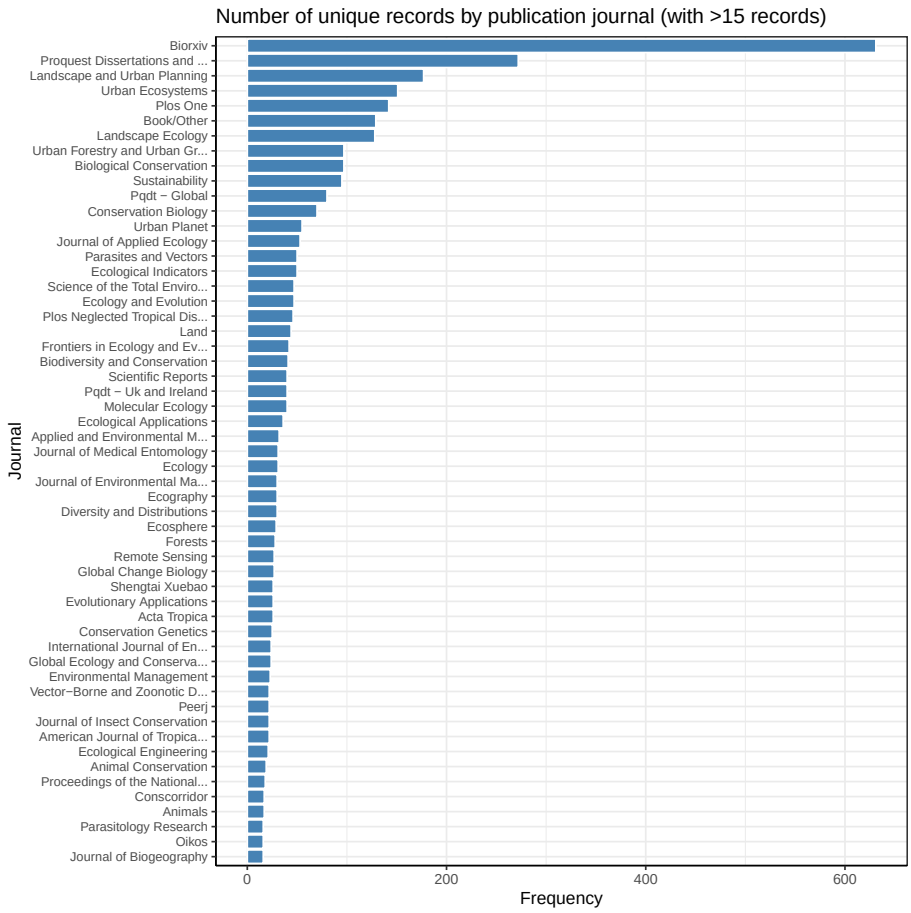


Figure 2.2: Unique record counts by journal of publication. Journals with <15 records were removed from the plot for visual clarity.

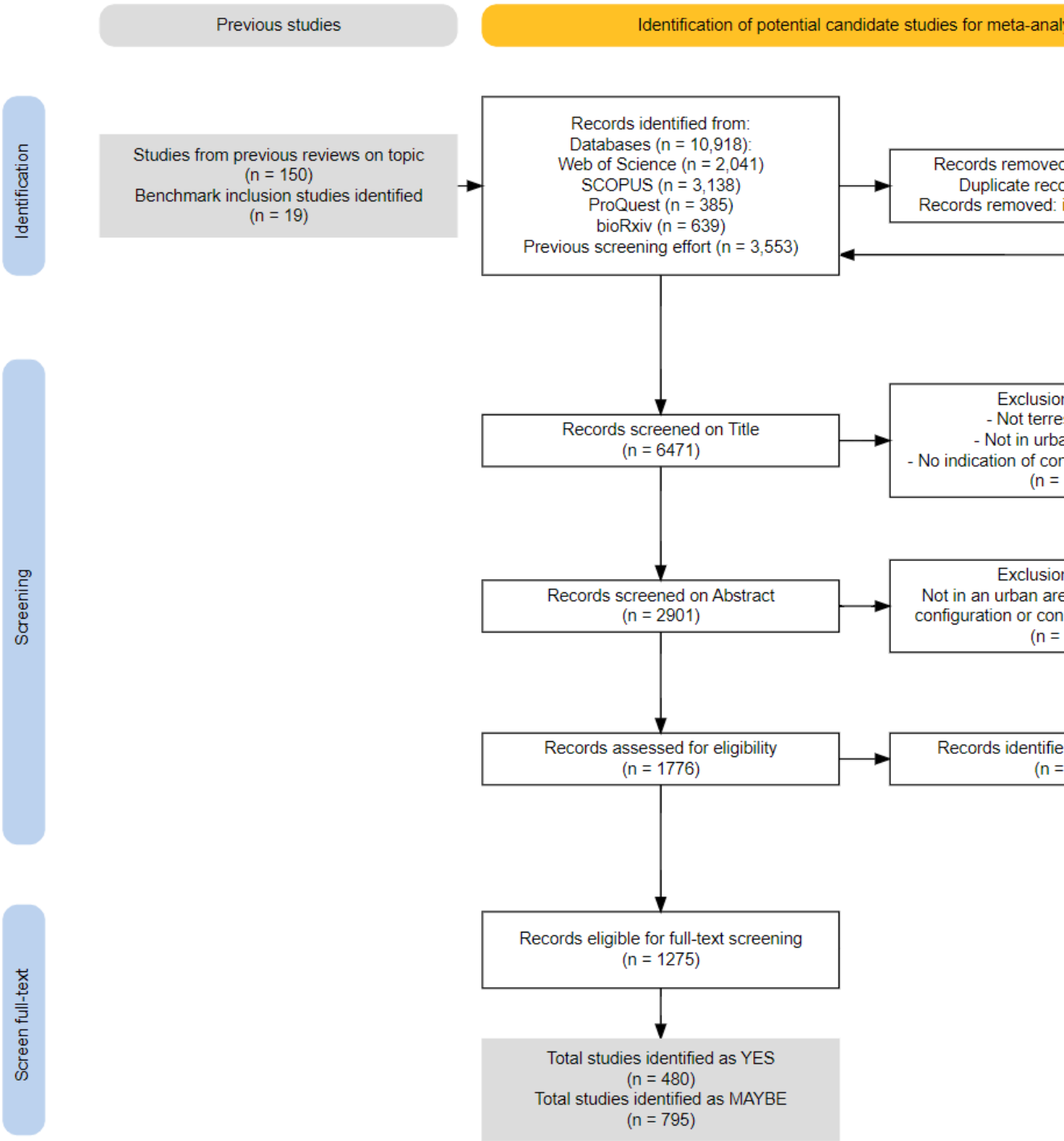


Figure 2.3: PRISMA diagram of Ti/Ab screening.

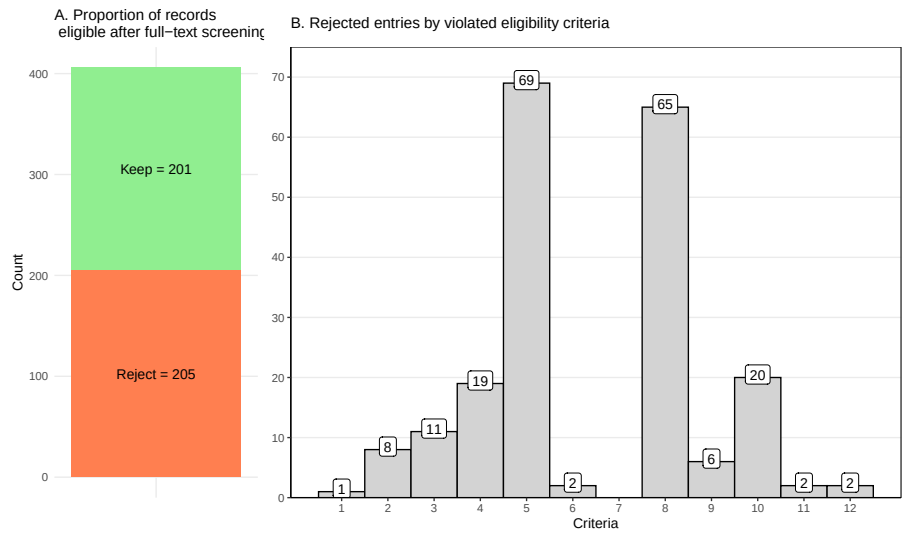


Figure 2.4: A: Full-text screening decisions for records identified as YES from the Ti/Ab screening. B: Rejected entries by violated eligibility criteria

2.4 Interactive map

We can also check which cities the studies were conducted in, and the proportion of the records identified as KEEP/REJECT.

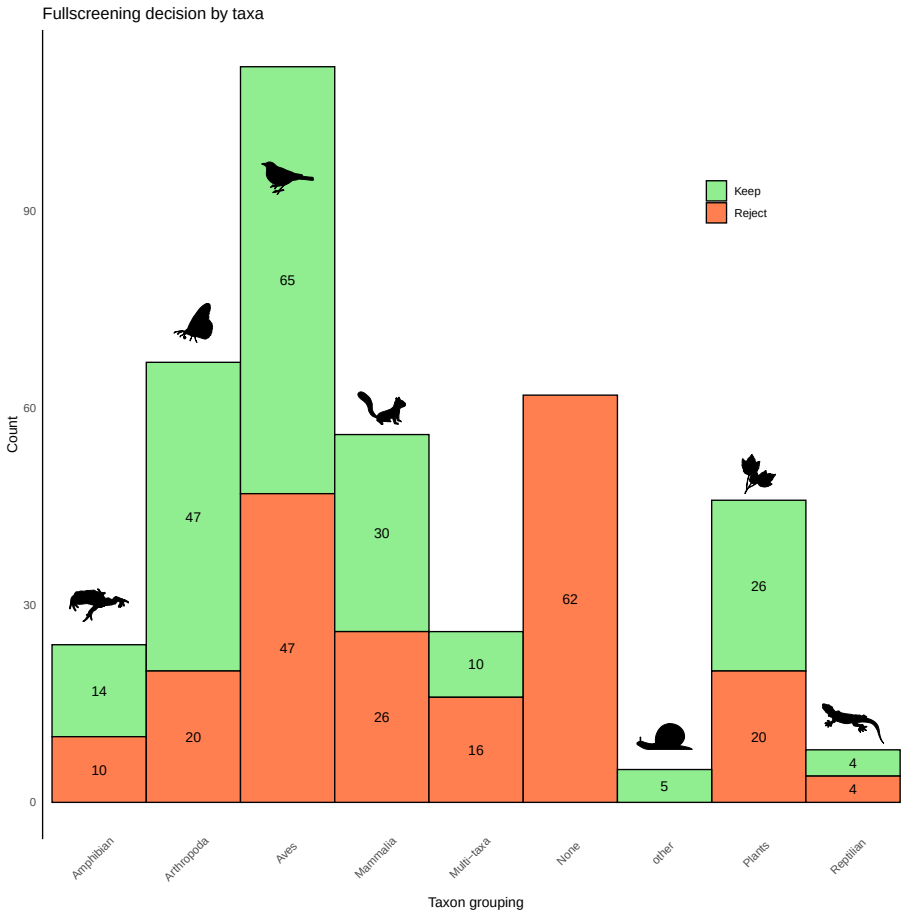
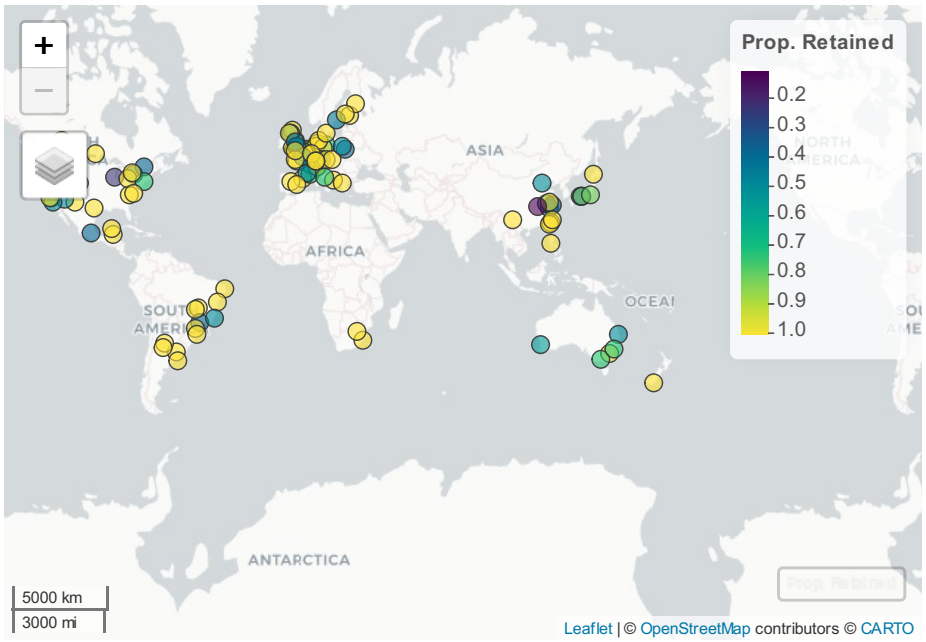


Figure 2.5: Full-text screening decisions by focal taxa of each potentially eligible record.



2.5 Data extraction

Despite soft efforts, I could not find reliable automated method of extracting relevant data from the pdfs. Currently, everything is being extracted manually, documented through zotero, using an extraction excel sheet I designed to enter data in a format that can easily be fed into the *metafor* (Viechtbauer 2010) R package to calculate standardized effect sizes from.

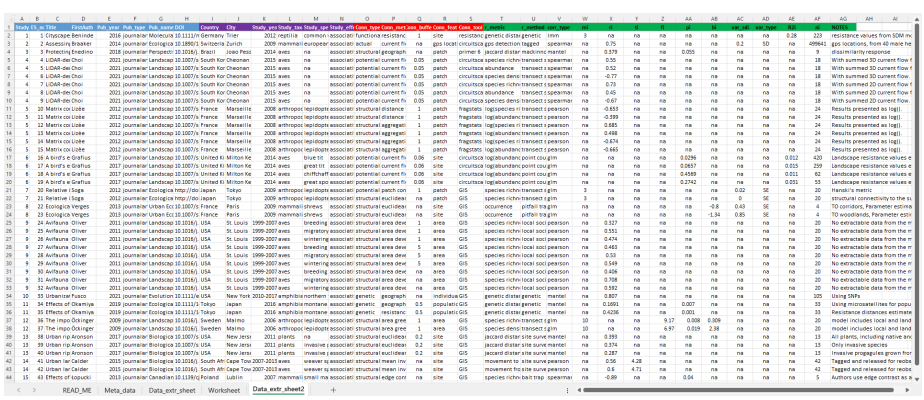


Figure 2.6: **Example extraction sheet with data extracted for several benchmark articles.**

2.6 Next steps:

- Finish writing up standardized protocol and submit to Environmental Evidence as a systematic review protocol.
- Finish full-screening MAYBEs for eligibility.
- Finish extracting relevant information for the rest of eligible papers.
- Calculate standardized effect sizes for each relevent effect.
- Conduct meta-regression on standardized effect sizes per connectivity metric and biodiversity metric that have a large enough sample size.
- Write-up analysis and insights.
- Get angry at R and cry a bunch.
- Publish meta-analysis.

Chapter 3

ebird checklist diversity in Canadian cities

The following R packages (and their versions) were used to generate this report.

```
##      leaflet        vegan      lattice      permute       ggpubr DataExplorer
##      "2.2.1"      "2.6-4"     "0.21-9"      "0.9-7"      "0.6.0"      "0.8.2"
##  tarchetypes      targets   kableExtra           sf      mapview    patchwork
##     "0.7.10"      "1.4.0"      "1.3.4"     "1.0-14"     "2.11.2"      "1.1.3"
##    rphylopic   synthesisr   CiteSource    lubridate      forcats      stringr
##      "1.2.2"      "0.3.0"      "0.0.1"      "1.9.3"      "1.0.0"      "1.5.0"
##        dplyr        purrr        readr        tidyr       tibble      ggplot2
##      "1.1.3"      "1.0.2"      "2.1.4"      "1.3.0"      "3.2.1"      "3.4.4"
##    tidyverse
##      "2.0.0"
```

3.1 Research question

> Does the built environment lower landscape connectivity for birds in the city, limiting dispersal and species richness in patches across the city? If so, what urban features most strongly contribute to this effect on species richness?

I plan to answer this research question using the ebird dataset and remote-sensed data to compare models of landscape connectivity for the most populated urban areas in Canada.

Here you'll find an overview of the framework I've developed to answer my research questions.

3.2 ebird dataset download

First, the dataset needs to be downloaded and cleaned up to get the relevant data to estimate species diversity. Note however, that the complete dataset as of Aug. 2023 is >500 gb of text (impossible to open with a text editor). Access requires use of a custom R package, *auk* (https://cornelllabofornithology.github.io/auk/).

The dataset itself is downloadable with a free account on the ebird website (https://ebird.org/home) or through Global biodiversity information facility (https://www.gbif.org/dataset/4fa7b334-ce0d-4e88-aaae-2e0c138d049e)

eBird and other community-collected datasets are generally "semi-structured" checklists where individuals can report both incidental observations, and complete surveys. To estimate biodiversity, the data needs to be filtered to minimize biases in the metrics of interest (species richness, eveness, etc.).

After initial data tidying and filtering, the dataset can be filtered using including geographic shapefiles to extract research-quality data by region of interest, i.e. Canadian urban areas.

3.3 Methods

Due to the enormity of the complete dataset, I needed a systematic way of filtering the dataset that was semi-automated, transparent, repeatable and not reliant on computationally intense for-loops. Using the *targets* R package (https://docs.ropensci.org/targets/), an open-source pipeline tool, and a collection of custom functions I've written, I developed the following pipeline to:

- Filter the original ebird dataset to the latest Canadian census data urban area delination for cities with >100000 people (2021, https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/index-eng.cfm), buffered by 5km to capture any data just outside the city limits.
- Format and tidy datasets to calculate biodiversity estimates for complete checklists.
- Calculate some biodiversity metrics (Right now only species richness, shannon diversity, evenness; other estimates as well as city-wide estimates can be added to the pipeline)

Each of these steps can be updated, changed, or branched from, and the whole pipeline rerun, with only the affected downstream objects needing reanalysis. This framework makes it easy to repeat the analysis for each city that is included without repeating the time-consuming steps for each city independently.

Below I provide an overview what the dataset looks like for only one city.

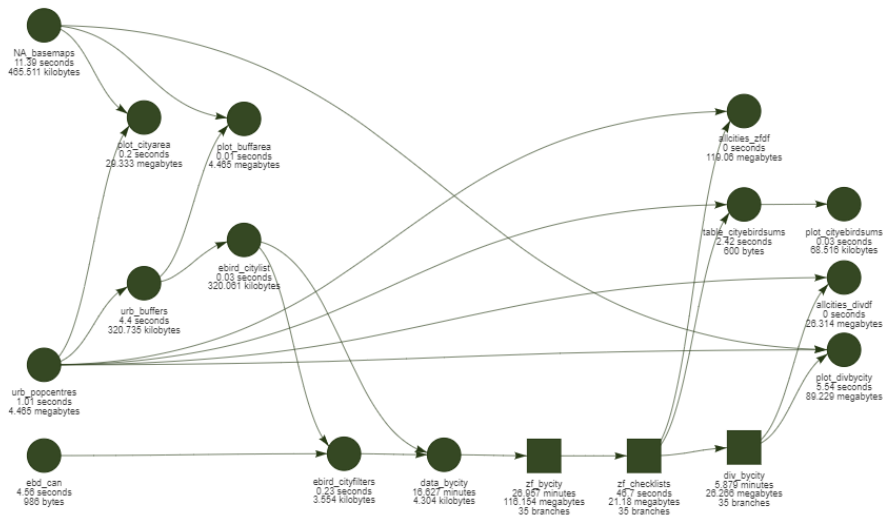


Figure 3.1: **Targets workflow network**. *Circles are R objects, and squares are objects that are iterated over for each subset of datapoints for the each city.*

3.4 Ottawa - Gatineau

Here we can see the number of unique, complete checklists for Ottawa - Gatineau by year. As you can see, the pandemic was a huge boon for urban birders, and the number of checklists *in* the city nearly doubled!

After some dataset tidying, I calculated biodiversity estimates for each of the checklists. *Note* this is super rough, and does not take into consideration checklist sampling effort or locations across the city.

Below is a series of violin plots, showing the distribution of the diversity metric across checklists for a given year.

Here you can see a summary (n# of checklists, median value, inter-quartile range) of the estimated biodiversity metrics for each year of complete checklists in the Ottawa - Gatineau urban area. Looks pretty consistent across years, which suggests that there's no immediately obvious source of bias in the dataset.

We can also take a look at the effort data for the checklists, to provides more information on the *potential* biases could come from different sampling methods.

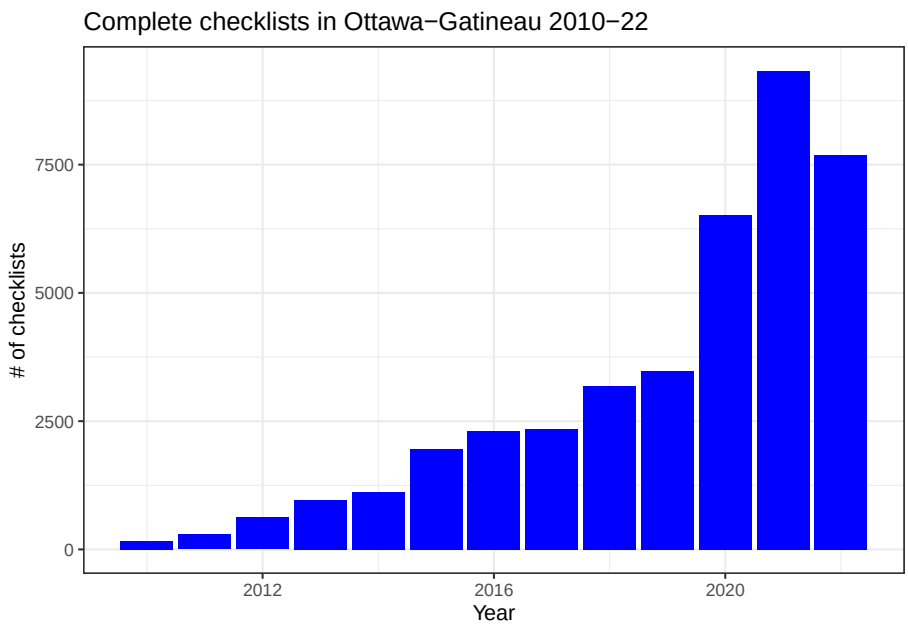


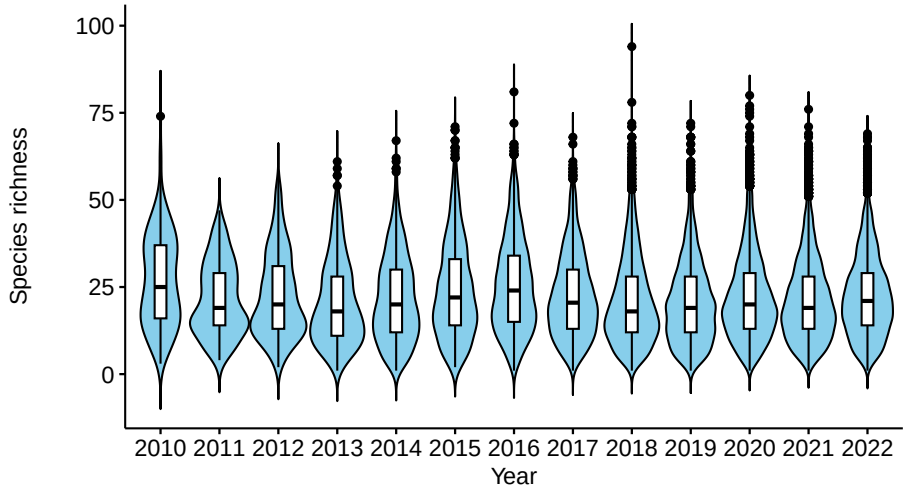
Figure 3.2: Number of complete eBird checklists in Ottawa - Gatineau from 2010 to 2022

3.5 Interactive map

Species richness estimates for complete eBird checklists in Ottawa - Gatineau for 2010 (left) and 2020 (right). Note the maps are synced together.

3.6 Next steps:

- Check checklist level habitat and land cover variables in area around each points. Select points and control for habitat amount, so that when I get to the modelling we can account for the effects of habitat amount on the relationship between the variables.
- Write scripts to incorporate landscapemetrics package into workflow.
- Estimate structural connectivity from landscape metrics (distance between checklists).
- Develope connectivity modelling that incorporate species informations.
- Be a cool scientist and write up stuff.
- Have a career as a cool scientist???



	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
n	159	263	535	716	898	1645	2083	2224	2957	3187	6083	8811	7266
median	25	19	20	18	20	22	24	20	18	19	20	19	21
iqr	21	15	18	17	18	19	19	17	16	16	16	15	15

Figure 3.3: Species richness estimates from eBird checklists for Ottawa - Gatineau for each year from 2010 - 2022

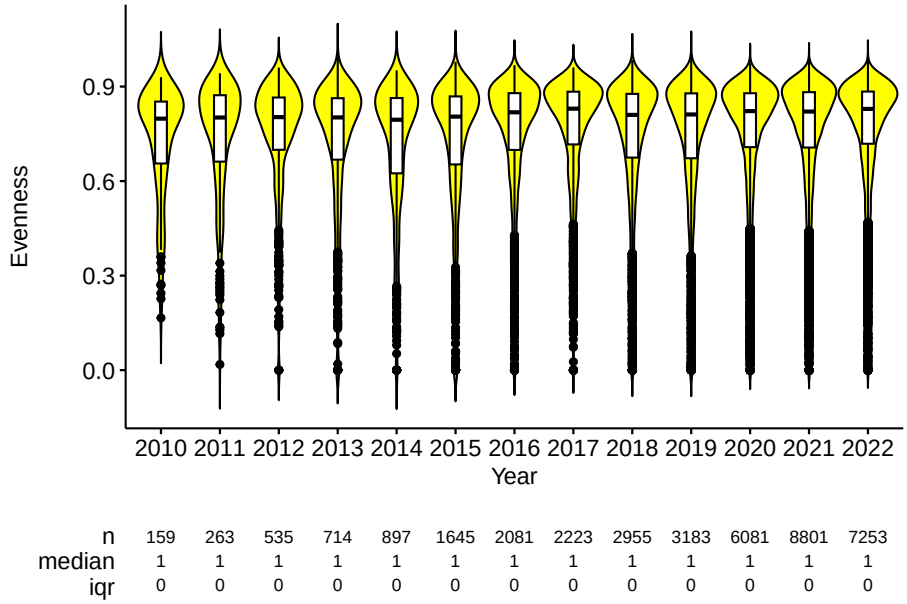


Figure 3.4: Species evenness estimates from eBird checklists for Ottawa - Gatineau for each year from 2010 - 2022

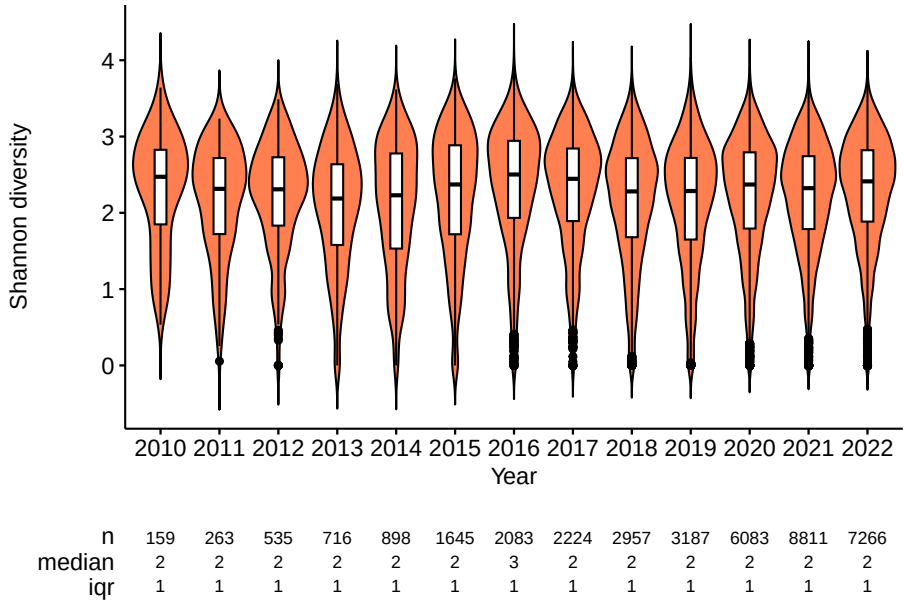


Figure 3.5: Shannon diversity estimates from eBird checklists for Ottawa - Gatineau for each year from 2010 - 2022

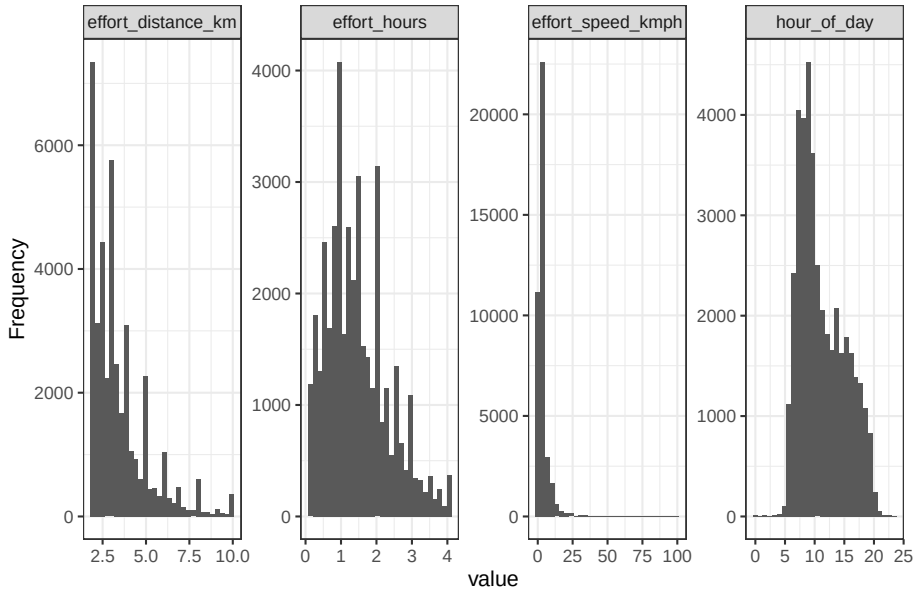


Figure 3.6: Overview of sampling effort for checklists

Chapter 4

ebfly data in canadian cities

The following R packages (and their versions) were used to generate this report.

```
##    ggspatial      geodata        terra     leafsync      leaflet        vegan
##      "1.1.9"      "0.5-9"     "1.7-55"      "0.1.0"      "2.2.1"      "2.6-4"
##      lattice      permute       ggpubr DataExplorer  tarchetypes      targets
##     "0.21-9"      "0.9-7"      "0.6.0"      "0.8.2"     "0.7.10"      "1.4.0"
##   kableExtra           sf      mapview    patchwork    rphylopic   synthesisr
##      "1.3.4"     "1.0-14"     "2.11.2"      "1.1.3"      "1.2.2"      "0.3.0"
##   CiteSource    lubridate      forcats      stringr        dplyr        purrr
##      "0.0.1"      "1.9.3"      "1.0.0"      "1.5.0"      "1.1.3"      "1.0.2"
##        readr        tidyr       tibble      ggplot2    tidyverse
##      "2.1.4"      "1.3.0"      "3.2.1"      "3.4.4"      "2.0.0"
```

4.1 ebutterfly data download

The most recent version of the dataset was downloaded from the Global Biodiversity Information Facility (https://www.gbif.org/dataset/cf3bdc30-370c-48d3-8fff-b587a39d72d6) . Note that downloading the data using the scripts on the github page (https://github.com/drewhabrich/Urban_eBfly) requires an account on the gbif website.

After removing incidental and museum records, checking for complete checklists by province/state shows an uneven distribution checklists in the dataset.

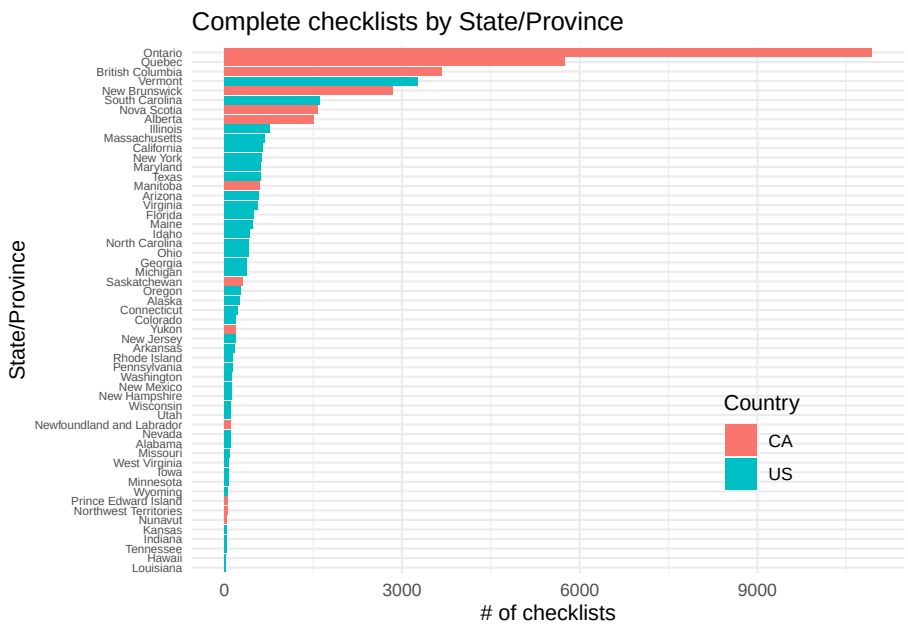


Figure 4.1: Number of complete eButterfly checklists by State/Province

4.2 Checklists filtered by geography

Using the latest Canadian census data (2021, https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/index-eng.cfm) filtered by urban area delination with >50000 people, buffered by 5km to capture any data just outside the city limits.

The vertical black line is at 100 checklists, which may be the lower limit to get accurate estimates of species diversity (I need to investigate species accumulation curves to see the comprehensiveness of the surveys). As you can see, the distribution of effort is not uniformly distributed across cities in Canada.

4.3 Checklists by year of submission

It is also important to consider the years that the checklists were submitted, in order to find remote-sensing imagery that matches temporal and spatial scale for landscape connectivity modelling. Here we can see the number of checklists by province as well.

Fairly uneven distribution across years, so maybe its better to limit analysis to years with sufficient data?

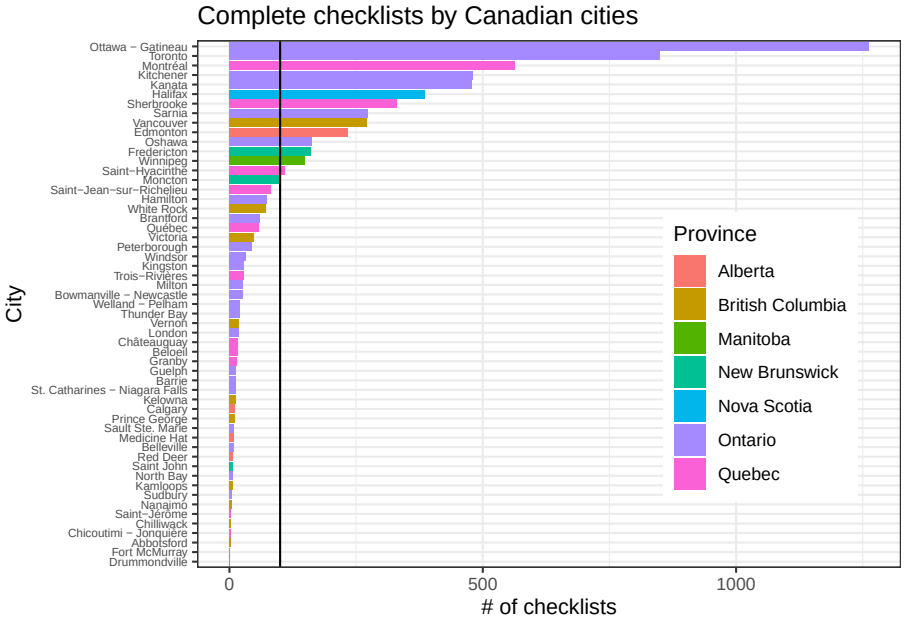
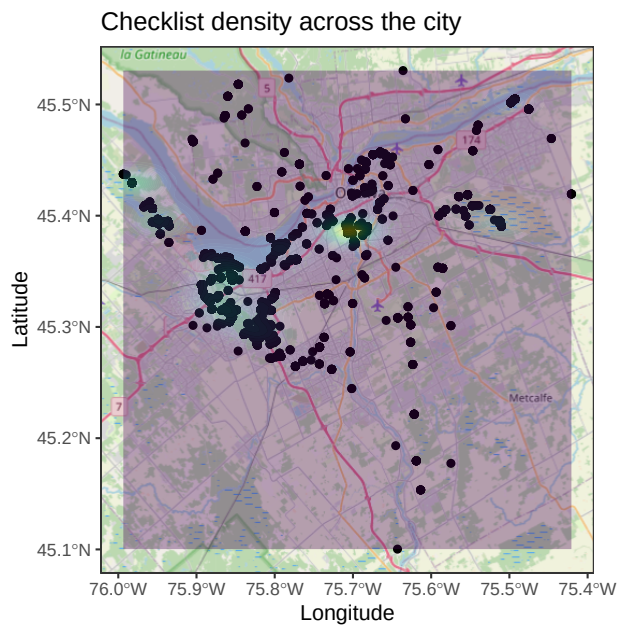
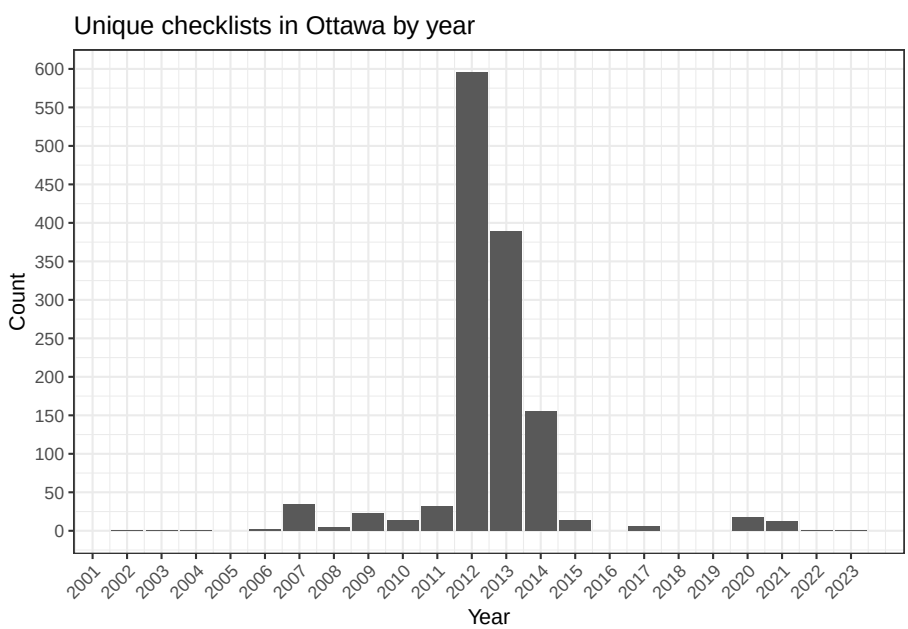


Figure 4.2: # of complete eButterfly checklists in the most populated Canadian cities coloured by Province.

4.4 Ottawa - Gatineau

Let's take a closer look at the single city with the most number of checklists:


Checklist density across the city
la Gatineau
Metcalfe
45.5°N
45.4°N
45.3°N
45.2°N
45.1°N
Latitude
76.0°W
75.9°W
75.8°W
75.7°W
75.6°W
75.5°W
75.4°W
Longitude


Unique checklists in Ottawa by year
Count
600
550
500
450
400
350
300
250
200
150
100
50
0
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
Year

4.5 Interactive map

You can interactively browse the dataset here, I've provided 2 layers to visualize the distribution of checklists by 1) sampling protocol, 2) year of checklist submission. You can turn them on and off with the layers button on the left (the square button with the image of stacked squares).

Clicking on one of the points will display the attributes for the checklist.

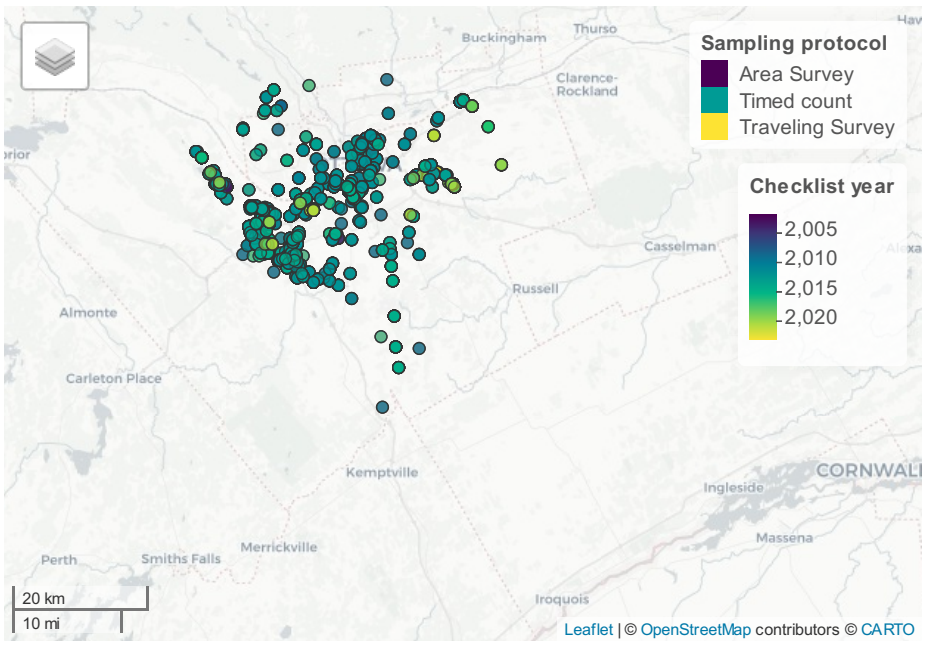


Figure 4.3: Interactive map of eButterfly checklists in Ottawa - Gatineau

4.6 Next steps:

- Check to see if there is ebutterfly data for cities in the US (probably just Vermont or a few states)
- Calculate biodiversity estimates
- Estimate structural connectivity from landscape metrics (distance between checklists)
- Connectivity modelling
- ???

Convention on Biological Diversity. 2022. "Kunming-Montreal Global Biodiversity Framework." https://prod.drupal.www.infra.cbd.int/sites/default/files/2022-12/221222-CBD-PressRelease-COP15-Final.pdf.

Correa Ayram, Camilo A., Manuel E. Mendoza, Andrés Etter, and Diego R. Pérez Salicrup. 2016. "Habitat Connectivity in Biodiversity Conservation: A Review of Recent Studies and Applications." *Progress in Physical Geography: Earth and Environment* 40 (1): 7–37. https://doi.org/10.1177/0309133315598713.

Fahrig, Lenore, Víctor Arroyo-Rodríguez, Eliana Cazetta, Adam Ford, Jill Lancaster, and Thomas Ranius. 2021. "Landscape Connectivity." In, 6788. Routledge.

Fletcher, Robert J., Jorge A. Sefair, Chao Wang, Caroline L. Poli, Thomas A. H. Smith, Emilio M. Bruna, Robert D. Holt, Michael Barfield, Andrew J. Marx, and Miguel A. Acevedo. 2019. "Towards a Unified Framework for Connectivity That Disentangles Movement and Mortality in Space and Time." Edited by Fangliang He. *Ecology Letters* 22 (10): 1680–89. https://doi.org/10.1111/ele.13333.

Grames, Eliza M., Andrew N. Stillman, Morgan W. Tingley, and Chris S. Elphick. 2019. "An Automated Approach to Identifying Search Terms for Systematic Reviews Using Keyword Co-Occurrence Networks." *Methods in Ecology and Evolution* 10 (10): 1645–54. https://doi.org/10.1111/2041-210X.13268.

Haddaway, N. R., M. J. Grainger, and C. T. Gray. 2021. *Citationchaser: An r Package for Forward and Backward Citations Chasing in Academic Searching*. https://doi.org/10.5281/zenodo.4543513.

Keeley, Annika T. H., Paul Beier, and Jeff S. Jenness. 2021. "Connectivity Metrics for Conservation Planning and Monitoring." *Biological Conservation* 255 (March): 109008. https://doi.org/10.1016/j.biocon.2021.109008.

LaPoint, Scott D., Niko Balkenhol, James D. Hale, Jonathan P. Sadler, and Rodney van der Ree. 2015. "Ecological Connectivity Research in Urban Areas." *Functional Ecology* 29 (7): 868–78. https://doi.org/10.1111/1365-2435.12489.

Lookingbill, Todd R., Emily S. Minor, Charles S. Mullis, Gabriela C. Nunez-Mir, and Philip Johnson. 2022. "Connectivity in the Urban Landscape (2015–2020): Who? Where? What? When? Why? And How?" *Current Landscape Ecology Reports* 7: 1–14. https://doi.org/10.1007/s40823-021-00068-x.

Nilon, Charles H., Myla F. J. Aronson, Sarel S. Cilliers, Cynnamon Dobbs, Lauren J. Frazee, Mark A. Goddard, Karen M. O'Neill, et al. 2017. "Planning for the Future of Urban Biodiversity: A Global Review of City-Scale Initiatives." *BioScience* 67 (4): 332–42. https://doi.org/10.1093/biosci/bix012.

Rudnick, Doborah, Sadie J Ryan, Paul Beier, Samual A Cushman, Fred Dieffenbach, Clinton Epps, Leah R Gerber, et al. 2012. "The Role of Landscape Connectivity in Planning and Implementing Conservation and Restoration Priorities." https://www.fs.usda.gov/research/treesearch/42229.

Taylor, Philip D., Lenore Fahrig, Kringen Henein, and Gray Merriam. 1993. "Connectivity Is a Vital Element of Landscape Structure." *Oikos* 68 (3): 571–73. https://doi.org/10.2307/3544927.

Tischendorf, Lutz, and Lenore Fahrig. 2000. "On the Usage and Measurement

of Landscape Connectivity." *Oikos* 90 (1): 7–19. https://doi.org/10.1034/j.1600-0706.2000.900102.x.

Viechtbauer, Wolfgang. 2010. "Conducting Meta-Analyses in r with the Metafor Package." *Journal of Statistical Software* 36 (3). https://doi.org/10.18637/jss.v036.i03.

Westgate, Martin. 2019. "Revtools: An R Package to Support Article Screening for Evidence Synthesis." *Research Synthesis Methods* 10 (4): 12. https://doi.org/10.1002/jrsm.1374.